

Synthetic Data for Helios Query & Response Testing

Data Policy and Standards Division

December 2024

Instructions for Querying Synthetic Data

Accessing Synthetic Data

There are two options for testing query and response of the synthetic data developed by CDC OPHDST.

Option 1: [Query Connector](#)

Query Connector Data Usage Policy

This site is for demo purposes only. Please do not enter PII on this website.

1 Enter patient info 2 Select patient 3 Select query 4 View patient record

Step 1: Enter patient information

Enter patient information below to search for a patient. We will query the connected network to find matching records.

The demo site uses synthetic data to provide examples of possible queries that you can make with the Query Connector. Select a query use case, a sample patient, and then click "fill fields" below.

[Fill fields](#) [Advanced](#)

Name

First Name Last Name

Samantha Augusto

Note: currently, not all resources are available through Query Connector.

Option 2: [Open FHIR Server endpoint URL](#)

1. <https://gw.interop.community/CDCOPHDSTConnect/open/Patient?name=Augusto> Save Share

GET 1. <https://gw.interop.community/CDCOPHDSTConnect/open/Patient?name=Augusto> Send

Params Authorization Headers (5) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
☒ name	Augusto		
Key	Value	Description	

Body Cookies Headers (25) Test Results 200 OK 460 ms 106.55 KB

Pretty Raw Preview Visualize HTML

```
1 <!DOCTYPE html>
2 <html lang="en-US" nonce="+4vCH8PqdemtSU+5S/wfUShIlepn0+XRks+Pbx/83v83VG3f">
3
4 <head>
5   <meta charset="UTF-8">
6   <title>Postman</title>
7
8   <!-- New Relic Integration -->
9
10  <script type="text/javascript" nonce="+4vCH8PqdemtSU+5S/wfUShIlepn0+XRks+Pbx/83v83VG3f">
11    ;window.NREUM||(NREUM=[]);NREUM.init={distributed_tracing:{enabled:true,
    cors_use_newrelic_header:false,cors_use_tracecontext_headers:true,allowed_origins:['https://
    bifrost-public-https-v4.gw.postman.com','https://bifrost-web-https-v4.gw.postman.co','https://
    bifrost-web-https-v4.gw.postman.com']},privacy:{cookies_enabled:false},ajax:{deny_list:['ban.
```

Note: requires an API tool.

Accessing Synthetic Data via Query Connector (1/4)

Option 1: [Query Connector](#)

All synthetic data (FHIR JSON format) is loaded into “OPHDST CDC Connectathon” Meld sandbox. This sandbox/endpoint is connected through **Query Connector**.

Step 1: Navigate to [Query Connector](#)

Step 2: Select "Advanced"

Step 3: Under FHIR Server (QHIN), select "OPHDST Meld: Direct"

Step 4: Type name of patient you want to query

Accessing Synthetic Data via Query Connector (2/4)

Option 1: Query Connector

Step 5: Select Patient

Query Connector

[Data Usage Policy](#)

i This site is for demo purposes only. Please do not enter PII on this website.

1

2

3

4

Enter patient info

Select patient

Select query

View patient record

[← Return to Enter patient info](#)

Step 2: Select a patient

The following records match your search. Select a patient to continue.

Name	DOB	Contact	Address	MRN	Actions
Samantha L Agosto	2001-02-20		22 Jump Street Springfield, GA	540697	Select patient

Accessing Synthetic Data via Query Connector (3/4)

Option 1: Query Connector

Step 6: Select condition out of list

Query Connector

Data Usage Policy

This site is for demo purposes only. Please do not enter PII on this website.

1

2

3

4

Enter patient info

Select patient

Select query

View patient record

[Return to Select patient](#)

Step 3: Select a query

We will request all data related to your selected patient and query. By only showing relevant data for your query, we decrease the burden on our systems and protect patient privacy. If you would like to customize the query response, click on the "customize query" button.

Query

Select query

Customize query

Advanced...

Submit

Accessing Synthetic Data via Query Connector (4/4)

Option 1: [Query Connector](#)

Step 7: View patient record

Note: Patient record will only include some demographic information and some observations at this time.

To query all available data fields, reference [Option 2](#).

The screenshot displays a web interface for a 'Patient Record'. At the top, the title 'Patient Record' is followed by the query 'Query: Syphilis case investigation'. On the left, a sidebar contains three tabs: 'Patient Info' (selected), 'Demographics', and 'Observations'. The main content area is divided into two sections. The top section, 'Patient Info', contains a form with the following fields: Patient Name (Samantha L. Agosto), DOB (02/20/2001), Current Age (23), Sex (female), Race (White), Ethnicity (Hispanic or Latino), Tribal Affiliation, Preferred Language, Address (22 Jump Street, Springfield, GA), Contact, and Patient Identifiers (Medical Record Number: 540697). The bottom section, 'Observations', is currently empty.

Patient Info	
Demographics	
Patient Name	Samantha L. Agosto
DOB	02/20/2001
Current Age	23
Sex	female
Race	White
Ethnicity	Hispanic or Latino
Tribal Affiliation	
Preferred Language	
Address	22 Jump Street Springfield, GA
Contact	
Patient Identifiers	Medical Record Number: 540697

Observations	
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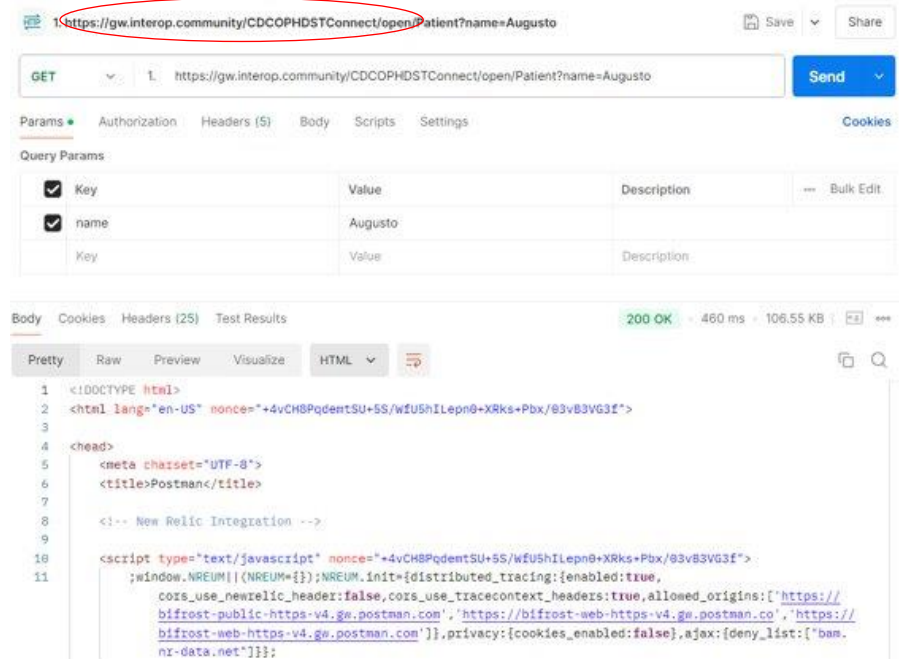
Accessing Synthetic Data via Open FHIR Server endpoint URL

Option 2: Open FHIR Server endpoint URL

Query any resource by referencing the open FHIR server URL via an API tool (e.g. *Postman*) (Fig 1.).

Open FHIR server URL:
<https://gw.interop.community/CDCOPHDSTConnect/open>

Fig 1. Example of query using API tool, Postman



Synthetic Data: Sexually Transmitted Illnesses

Sexually Transmitted Illnesses: Syphilis

Samantha Agosto, 23 y/o female, presents to a hospital in Springfield, GA with fatigue and a rash on multiple body parts. She is found positive for *Treponema pallidum* by nucleic acid amplification test (NAAT) from a genital specimen and is prescribed Amoxicillin/ Clavulanate to be taken orally.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Samantha L Agosto
DOB	2001-02-20
Race	White
Ethnicity	Hispanic or Latino
Deceased Indicator	No
Pregnancy (SNOMED)	Not pregnant (finding)

Data Element	Example
Medication Description	Amoxicillin 800 MG / clavulanate 125 MG Oral Tablet
Medication Code	393512
Problem Description	Primary genital syphilis
Problem Code (ICD-10)	A51.0
Lab Test Code	91846-6
Observation Value	260373001
Lab Test Date	6/12/2024

Sexually Transmitted Illnesses: Gonorrhea

Erica Baker, 54 y/o female, presents to a hospital in Jenniferburgh, MD due to a persistent fever, painful urination, and sharp abdomen pain. The provider performs an NAAT test on a urethra sample and finds it to be positive for *Neisseria gonorrhoeae*. She is treated with a single intramuscular injection of ceftriaxone.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Erica L Baker
DOB	1970-09-24
Race	American Indian or Alaska Native
Pregnancy	Not pregnant (finding)
Pregnancy (SNOMED)	60001007
Social History	Heavy smoker (over 20 per day)
Social History (SNOMED)	56771006

Data Element	Example
Provider Name	Jesse Kirk
Encounter Effective Time Low	2/2/2024 1:58:54 AM
Encounter Type	CPT - 99213
Encounter Reason Code	236682002
Encounter Reason Display	Gonococcal urethritis (disorder)
Condition Category Code	encounter-diagnosis

Sexually Transmitted Illnesses: Chlamydia

Peter Chamberlain, 57 y/o male, presents to a hospital in Pleasantville, AR with frequent and painful urination and testicular pain. He is found positive for chlamydia trachomatis infection by antibody test. He is treated with a single dose of azithromycin taken orally.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Peter J Chamberlain
DOB	1967-09-14
Race	White
Ethnicity	Not Hispanic or Latino
US Core Code (Ethnicity)	2186-5
Social History	Multiple Current Sexual Partners (finding)
Social History (SNOMED)	228460004

Data Element	Example
Facility Name	Hospital B
Encounter Code	AMB
Encounter reason Display	Chlamydia trachomatis infection
Lab Test Code	31768-5
Lab Test Description	Chlamydia trachomatis Ag [Presence] in Blood

Sexually Transmitted Illnesses: Herpes

Timothy Pilcow, 23 y/o female, presents to a hospital in Springfield, GA with small sores and genital pain. She is tested for Herpes Simplex 1 + 2 via NAAT by probe detection and is found positive. To treat her symptoms, she is administered famciclovir orally.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Timothy W Pilcow
DOB	2001-02-22
Observation Issued	2024-06-26T13:23:44.000-04:00
Patient Street Address	24 Jump Street
Social History	Multiple Current Sexual Partners (finding)
Social History (SNOMED)	228460004

Data Element	Example
Lab Test Code	20444-6
Lab Test Code System	LOINC
Lab Test Description	Herpes simplex virus 1+2 DNA [Presence] in Specimen by NAA with probe detection
Observation Interpretation Code	POS
Date/Time of Analysis	6/15/2024 9:28:14 PM

Synthetic Data: Newborn Screenings

Newborn Screening: Hypoplastic Left Heart Syndrome

Emily Johnson, was born in Springfield, IL in July 2024. The next month, her mother brought her back to the hospital with blueish skin, lips, and a rapid heartbeat. She was found with low blood oxygen levels and diagnosed with Hypoplastic left heart syndrome. Emily was treated with an injection of alprostadil and dopamine hydrochloride.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Patient Name (First, M, Last)	Emily R Johnson
DOB	2024-07-15
Guardian Name (First, M, Last)	Roselyn J Johnson
Relationship to Patient	Next-of-Kin
Encounter Effective Time Low	2024-08-20 17:55:40
Symptom Display	Neonatal respiratory distress
Symptom Code	1179627006

Data Element	Example
Symptom Display (1)	Neonatal tachycardia (disorder)
Symptom Code (1)	275521001
Lab Test Date	2024-08-23
Observation Category Code	vital-signs
Concentration Code	3150
Concentration Display	Inhaled oxygen concentration
Observation value: Concentration	84

Newborn Screening: Infant Hearing Loss

James Martinez was born in Riverbend, CO in August 2024. His hearing was tested and he was found to have normal hearing in his left ear and hearing loss in his right ear. He was referred to a specialist for this issue.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Patient Name (First, M, Last)	James P Martinez
DOB	2024-08-08
Race	Black or African American
Guardian Name (F, M, L)	Cole T Martinez
Facility Name	Hospital H
Patient Name (F, M, L)	James P Martinez

Data Element	Example
Symptom Code	425091001
Symptom Display	Difficulty hearing normal speech tones (finding)
Symptom Code (1)	441719005
Symptom Display (1)	Speech and language developmental delay due to hearing loss (disorder)
Encounter Reason Display	Speech and language developmental delay due to hearing loss (disorder)

Synthetic Data: Respiratory Conditions

Respiratory Conditions: Covid-19

Lindsay Luke, 41 y/o female, admits to a hospital in St. Mounds, OK with fever, dyspnea, and nausea. She is found positive for SARS-CoV-2 by nucleic acid amplification test (NAAT) from a throat specimen and is diagnosed with Acute Respiratory Distress Syndrome. She is prescribed Paxlovid to be taken orally.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Lindsay H Luke
DOB	1983-07-23
Race	White
Ethnicity	Not Hispanic or Latino
Deceased Indicator	No
Pregnancy (SNOMED)	Not pregnant (finding)

Data Element	Example
Medication Code	2599543
Problem Description	Acute respiratory distress syndrome due to disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)
Problem Code (SNOMED)	674814021000119000
Lab Test Code (SNOMED)	101289-7
Observation Value Code	260373001

Respiratory Conditions: Influenza

Anita Hogan, 31 y/o female, presents to a hospital in South Jeffrey, VT, with a fever and dry cough. The provider performs a culture from a nasopharyngeal swab and finds it to be positive for Influenza Type B. Anita is treated with oseltamivir to be taken orally.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Erica L Baker
DOB	1993-03-14
Race	White
Pregnancy	Not pregnant (finding)
Pregnancy (SNOMED)	60001007
Social History: Smoking	Non-smoker (finding)
Social History: Smoking (SNOMED)	8392000

Data Element	Example
Provider Name	George Duran
Encounter Effective Time Low	1/22/2024 5:33:32 PM
Encounter Type	CPT - 99213
Encounter Reason Code	24662006
Encounter Reason Display	Influenza caused by Influenza B virus (disorder)
Symptom Display	Fever caused by virus (finding)

Respiratory Conditions: Respiratory Syncytial Virus (RSV)

Pius Smith, 42 y/o male, presents to a hospital in Pamelaland, CT, with a sore throat, cough, and fever. He is found positive for RSV by nucleic acid amplification test (NAAT) from a Nasopharynx probe and is treated with a single dose injection of Abrysvo.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	Pius C Smith
DOB	1982-06-15
Race	White
Ethnicity	Hispanic or Latino
US Core Code (Ethnicity)	2186-2
Facility Street Address	648 Jacqueline Parkways Apt. 438
Facility State	CT

Data Element	Example
Encounter Code	IMP
Encounter Type	CPT - 99213
Encounter reason Display	Encephalitis caused by human respiratory syncytial virus (disorder)
Lab Test Code	76089-2
Lab Test Code system	LOINC
Lab Test Description	Respiratory syncytial virus RNA [Presence] in Nasopharynx by NAA with probe detection

Respiratory Conditions: Tuberculosis

David Lee, 51 y/o male, presents to a hospital in Canterbury, MO with cough, fever, shortness of breath, and chest pain. He is tested for *Mycobacterium tuberculosis* by NAAT test and is found positive. David is treated with a 300 MG oral tablet of Isoniazid.

The following fields are examples of what can be queried from the open FHIR Server endpoint URL (not an exhaustive list):

Data Element	Example
Name (First, M, Last)	David B Lee
DOB	1973-09-03
Observation Issued	2024-06-24T00:49:20.000-04:00
Patient Street Address	126 Johnson Turnpike
Social History: Smoke Status	Heavy smoker (over 20 per day) (finding)
Social History (SNOMED)	56771006

Data Element	Example
Lab Test Code	85362-2
Lab Test Code System	LOINC
Lab Test Description	Mycobacterium tuberculosis complex DNA [Presence] in Sputum or Bronchial by NAA with probe detection
Observation Interpretation Code	POS
Date/Time of Analysis	1/4/2024 8:51:22 AM
Lab Test Code	85362-2

Thank you!

For more information, contact CDC OPHDST
DSMH@CDC.GOV
1-800-CDC-INFO (232-4636)
TTY: 1-888-232-6348 [cdc.gov](https://www.cdc.gov)
Follow us on X (Twitter) @CDCgov & @CDCEnvironment

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the U. S. Centers for Disease Control and Prevention.

